NCBI Bookshelf. A service of the National Library of Medicine, National Institutes of Health.

StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-.

**This publication is provided for historical reference only and the information may be out of date.**

# Berry Aneurysm(Archived)

George A. Koutsothanasis; Raghuram Sampath.

Author Information and Affiliations

Last Update: September 26, 2022.



## Introduction

Berry (saccular) aneurysms are the most common type of intracranial aneurysm, accounting for approximately 90% of cerebral aneurysms. Generally speaking, ballooning occurs due to a weakened area in the wall of a blood vessel in the brain. Depending on the size of the aneurysm, their symptomatology ranges from asymptomatic to intracerebral hemorrhage (subarachnoid) in the most extreme cases. Berry aneurysm is an older term that has been largely replaced by saccular aneurysm (see **Image:** Berry Aneurysm). The new nomenclature goes against the tradition of likening a pathologic process to various fruit products. Aneurysm is from the Greek word aneurysma, meaning dilation.

## Etiology

**Genetic Factors**

- Connective tissue disorders that weaken artery walls
- Polycystic kidney disease
- Arteriovenous malformations
- History of an aneurysm in 2 or more first-degree family members

**Other Factors**

- Untreated high blood pressure
- Cigarette smoking
- Drug abuse, including cocaine and amphetamines, which, via their toxidrome, increases blood pressure significantly. Intravenous drug abuse can cause infectious mycotic aneurysms.
- Heavy alcohol intake
- Heavy caffeine intake

**Less Common**

- Head trauma
- Infection in the arterial wall from bacteremia - mycotic aneurysms

**Risk Factors for Aneurysm Rupture**

- **Smoking:** Not only can smoking promote the development of cerebral aneurysms, but it can also lead to their growth and rupture.
- **High blood pressure:** Chronic damage to arteries can lead to weakness, potentially forming an aneurysm and increasing the likelihood of rupture.
- **Size:** Larger aneurysms rupture more frequently than smaller ones.
- **Location:** Posterior circulation, including the posterior communicating artery (PCOM), the posterior cerebral artery (PCA), and the vertebrobasilar artery, is more common
- **Growth:** If aneurysms grow during a surveillance period despite being small, they are at an increased risk of rupture.
- **Family history:** A family history of rupture increases the risk of rupture.
- Irregular shape, multilobed, daughter sac, blebs

Those with previous ruptures or intracranial bleeds are at the highest risk of cerebral artery aneurysm rupture.[1]

## Epidemiology

The epidemiology of patients with berry aneurysms includes:

- Cerebral aneurysms occur in 3 to 5 percent of the general population.
- The prevalence is 3.2 percent according to radiographic and autopsy series.[2]
- 0.7 to 1.9 percent rupture, causing subarachnoid hemorrhage (SAH) [3]
- The mean age is 50 years.
- More common in females
- Twenty to thirty percent of patients with aneurysms have multiple aneurysms.
- Eighty-five percent of aneurysms are located in the anterior circulation, mainly in the circle of Willis.

**Common Sites in the Anterior Circulation**

- Junction of the anterior communicating artery with the anterior cerebral artery
- Junction of the posterior communicating artery with the internal carotid artery
- Bifurcation of the middle cerebral artery

**Common Sites in the Posterior Circulation**

- The top of the basilar artery, called the basilar apex
- Junction of the basilar artery and the superior or anterior inferior cerebellar arteries
- Junction of the vertebral artery and the posterior inferior cerebellar artery [4]

## Pathophysiology

Berry aneurysms are due to outpouchings in the blood vessel wall, which hereditary factors or acquired diseases can cause. Repetitive trauma and shearing forces against the weak point in the blood vessel's wall cause aneurysms to enlarge.[5]

Older dogma considers berry aneurysms to be passively enlarging vascular structures. More recent evidence suggests that berry aneurysms, along with other forms of aneurysms, are created and enlarged through continuous inflammation and tissue degradation.

## Histopathology

Some traditional classifications pose this as a model for organizing the types of intracranial aneurysms:

- **Mycotic:** An artery having sustained damage secondary to an infectious process. These aneurysms tend to be located more distally.
- **Luetic:** Also called syphilitic aneurysms, usually in the aorta and known as syphilitic aortitis, seen in the tertiary stage of syphilis
- **Arteriosclerotic:** These vessel widenings (aneurysms) occur due to arteriosclerosis, damaging the vessel wall [6]
- **Traumatic:** Vessels may be damaged or disrupted in the setting of trauma, both penetrating to the head and blunt injuries - these are commonly pseudoaneurysms.
    - These types of aneurysms represent less than 1% of cerebral aneurysms. However, they are prone to grow rapidly, leading to increased morbidity and mortality. Vigilance should be implemented to detect and treat these in the trauma setting.
- **Congenital:** Certain conditions predispose people to be born with aneurysms. One example is ADPKD (autosomal dominant polycystic kidney disease).[7]

## History and Physical

The clinical picture of berry aneurysms includes the most severe manifestation of a major aneurysmal rupture, such as a subarachnoid hemorrhage, to minor hemorrhage, also known as a sentinel bleed, nonhemorrhagic scenarios, or asymptomatic situations in which the aneurysm is found incidentally.[8]

Many of these patients may be brought in in extremis, so ABCs take priority. Assess the patient's airway and breathing and establish good IV access. In the trauma setting, prioritize your primary survey and ensure a thorough secondary survey is conducted. Do not be distracted by dramatic extremity injuries. Once the patient has been stabilized hemodynamically, prioritize emergent neuroimaging.

Other patients may present less extreme and warrant a thorough history and physical, even outside the setting of trauma.

Your History should include some key elements as described below:

- **Headache -** WHOML (worst headache of my life), thunderclap headache. The headache need not be the worst of the patient's life. Even a headache that is different from past headaches should raise one's suspicion of subarachnoid hemorrhage.
- **Change in level of consciousness:** Increased intracranial pressure decreases the perfusion pressures required to oxygenate the brain properly. Patients may appear confused or less alert than expected.
- Seizures are present in 25% of aneurysmal subarachnoid hemorrhages. They may be focal or generalized.
- **Meningeal signs -** Patients may exhibit classic meningeal irritation, complaining of neck pain or stiffness.
- Focal neurologic deficits include changes in strength, sensation, ability to speak or express themselves, and memory loss.
- **Visual symptoms-** blurry vision, double vision (diplopia), defects in the patient's visual field

The physical exam should include, but not be limited to, the following elements:

- A complete neurologic exam:
  - Cranial nerve testing
  - Strength of the upper and lower extremities and face
  - Sensations of the upper and lower extremities and face
  - Assess speech (dysarthric, disorganized, comprehensible)
  - Assess for meningismus, Kernig's, and Brudzinski's signs
- HEENT
  - Examine for signs of trauma to the head
    - obvious signs of injury, including lacerations, abrasions, contusions, skull depressions/fractures
    - Ears - look for blood behind the tympanic membrane (hemotympanum), assess for CSF otorrhea
    - Eyes - ensure PERRLA (pupils equal, round, and reactive to light and accommodation)
      - Look for other signs of eye trauma (iritis, conjunctival hemorrhage, globe rupture), which may indicate a severe mechanism of injury
    - Nose - assess for nasal fractures, CSF rhinorrhea [9]

## Evaluation

**Imaging**

- NCCT (non-contrast CT) of the head [10]
- With or without CT angiography of the head, especially if NCCT is negative
- MRI/MRA are also options, if available
- Formal cerebral catheter angiogram

**Lumbar Puncture**

- In cases with negative head CT, a strong family history of aneurysms, or ruptured aneurysms
- Consider after negative NCCT
- may aid in making an alternative diagnosis as the cause of headache (meningitis, idiopathic intracranial hypertension)
- Looking for **xanthochromia** and **RBC count**
- Xanthochromia - the CSF has a yellowish appearance since blood has been broken down enzymatically and is leaking bilirubin into the surrounding fluid [11]



## Treatment / Management

Medical management of these patients should begin in the emergency department. Within the first 24 hours, there is a high risk of rebleeding, which can be significantly reduced by controlling blood pressure. Systolic pressure should be kept at less than 140 mmHg.[12]

- Labetalol and nicardipine are good first-line agents.
- Nitroglycerin and nitroprusside should be avoided, as they are associated with an increased risk of increased intracranial pressure.

Remember to control the patient's pain. It is not only the right thing to do, but adequate pain control also makes it easier to control blood pressure.

Use antiemetics as needed. Nimodipine may be added in the first 96 hours to help decrease the degree of vasospasm.[13] Antiseizure medications should also be initiated.

## Differential Diagnosis

On the differential for saccular aneurysms, there are many disease processes that cause headaches:



- Headache (tension, migraine, cluster)
- Meningitis
- Encephalitis
- SOL (space-occupying lesions such as tumors)
- Pituitary masses (endocrine)
- Metabolic disturbances
- Cerebral venous thrombosis

- Idiopathic intracranial hypertension
- Temporal arteritis

## Prognosis

Many aneurysms never cause a patient any discomfort or sequelae. If an aneurysm does rupture, the prognosis depends on several factors, including:

- Prior neurologic conditions
- Age
- Comorbidities
- Aneurysm location
- Time from rupture to first contact with a doctor
- Bleeding extent or grade of subarachnoid hemorrhage and (rebleeding)
- Aneurysm treatment success
- Vasospasms in the ensuing days
- Development of hydrocephalus and seizures

Twenty-five percent of patients with ruptured aneurysms do not survive the first 24 hours. Of those who survive initially, 25 percent die from other complications in the next 6 months.[14]



## Complications

The complications that can manifest with berry aneurysms are as follows:

- Rebleeding
- Hyponatremia
- Vasospasm
- Hydrocephalus
- Seizures
- Cardiac stress

## Deterrence and Patient Education

Patients should know how to utilize the emergency medical services (EMS) available to them, such as emergency rooms, and when to seek the expert advice of an emergency physician if a severe or different headache develops from their baseline headaches.[15]

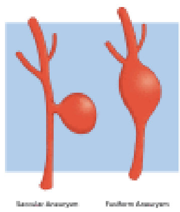


## Enhancing Healthcare Team Outcomes

Currently, research is being conducted in various fields, including the targeting of specific genes and the improvement of diagnostic accuracy in neuroimaging, which enables swift and accurate diagnosis. This, in turn, improves patient outcomes and reduces the morbidity and mortality associated with saccular aneurysms, particularly those that have ruptured. Outcomes can be enhanced by an interprofessional team, including EMS personnel, emergency department nurses and physicians, radiologists, neurologists, physiatrists, and therapists.

## Review Questions

- Access free multiple choice questions on this topic.
- Click here for a simplified version.
- Comment on this article.



**Figure**

Berry Aneurysm Illustrated by K Humphreys

## References


1. Alg VS, Sofat R, Houlden H, Werring DJ. Genetic risk factors for intracranial aneurysms: a meta-analysis in more than 116,000 individuals. Neurology. 2013 Jun 04;80(23):2154-65. [PMC free article: PMC3716358] [PubMed: 23733552]
2. Vlak MH, Algra A, Brandenburg R, Rinkel GJ. Prevalence of unruptured intracranial aneurysms, with emphasis on sex, age, comorbidity, country, and time period: a systematic review and meta-analysis. Lancet Neurol. 2011 Jul;10(7):626-36. [PubMed: 21641282]
3. Etminan N, Rinkel GJ. Unruptured intracranial aneurysms: development, rupture and preventive management. Nat Rev Neurol. 2017 Feb 01;13(2):126. [PubMed: 28145447]
4. Chalouhi N, Hoh BL, Hasan D. Review of cerebral aneurysm formation, growth, and rupture. Stroke. 2013 Dec;44(12):3613-22. [PubMed: 24130141]
5. Samuel N, Radovanovic I. Genetic basis of intracranial aneurysm formation and rupture: clinical implications in the postgenomic era. Neurosurg Focus. 2019 Jul 01;47(1):E10. [PubMed: 31261114]
6. Texakalidis P, Sweid A, Mouchtouris N, Peterson EC, Sioka C, Rangel-Castilla L, Reavey-Cantwell J, Jabbour P. Aneurysm Formation, Growth, and Rupture: The Biology and Physics of Cerebral Aneurysms. World Neurosurg. 2019 Oct;130:277-284. [PubMed: 31323409]
7. Yong-Zhong G, van Alphen HA. Pathogenesis and histopathology of saccular aneurysms: review of the literature. Neurol Res. 1990 Dec;12(4):249-55. [PubMed: 1982169]
8. Aktham A, AbdulAzeez MM, Hoz SS. Surgical Intervention of Intracerebral Hematoma Caused by Ruptured Middle Cerebral Artery Aneurysm in Neurosurgery Teaching Hospital, Baghdad, Iraq. Neurol India. 2020 Jan-Feb;68(1):124-131. [PubMed: 32129261]
9. CROMPTON MR. Intracerebral haematoma complicating ruptured cerebral berry aneurysm. J Neurol Neurosurg Psychiatry. 1962 Nov;25(4):378-86. [PMC free article:

PMC495503] [PubMed: 14023947]
10. Wu X, Chen X, Zhu J, Chen Q, Li Z, Lin A. Imaging detection of cerebral artery fenestrations and their clinical correlation with cerebrovascular diseases. Clin Imaging. 2020 Jun;62:57-62. [PubMed: 32066034]
11. Fisher CM, Roberson GH, Ojemann RG. Cerebral vasospasm with ruptured saccular aneurysm--the clinical manifestations. Neurosurgery. 1977 Nov-Dec;1(3):245-8. [PubMed: 615969]
12. Wang JW, Li CH, Tian YY, Li XY, Liu JF, Li H, Gao BL. Safety and efficacy of endovascular treatment of ruptured tiny cerebral aneurysms compared with ruptured larger aneurysms. Interv Neuroradiol. 2020 Jun;26(3):283-290. [PMC free article: PMC7254617] [PubMed: 31930939]
13. Nornes H, Magnaes B. Intracranial pressure in patients with ruptured saccular aneurysm. J Neurosurg. 1972 May;36(5):537-47. [PubMed: 5026540]
14. Wiebers DO, Whisnant JP, O'Fallon WM. The natural history of unruptured intracranial aneurysms. N Engl J Med. 1981 Mar 19;304(12):696-8. [PubMed: 7464862]
15. Wiebers DO, Whisnant JP, Sundt TM, O'Fallon WM. The significance of unruptured intracranial saccular aneurysms. J Neurosurg. 1987 Jan;66(1):23-9. [PubMed: 3783255]

**Disclosure:** George Koutsothanasis declares no relevant financial relationships with ineligible companies.

**Disclosure:** Raghuram Sampath declares no relevant financial relationships with ineligible companies.



Bookshelf ID: NBK557480 PMID: 32491412